Suchakra Kumar Gattu

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SUMMARY

AI Engineer building **One08labs**, CloudAngles' AI voice orchestration platform. Full-stack LLMOps experience gained building **mlangles** – spanning feature development, model training, and inference optimization through to technical leadership – with production depth in voice AI telephony, LLM inference, and frequent customer-facing product releases.

EDUCATION

Keshav Memorial Engineering College, Osmania University **Hyderabad**
B.E., Computer Science and Engineering (Specialization: AI & ML) *2021 – 2025*

- **Coursework:** Machine Learning, Advanced Machine Learning, Deep Learning, Operating Systems, Computer Networks, DBMS, Data Structures and Algorithms, Image Processing, Information Retrieval

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, C
AI / ML	PyTorch, TensorFlow, Transformers, LoRA, QLoRA, SFT, DPO, PEFT, RAG, LangChain, LangGraph
Voice AI	SIP, Telephony, STT, TTS, Voice Agents, WebRTC, LiveKit
Cloud & Containers	AWS (Amazon EC2, ECS, Bedrock, EBS, Security Groups), Docker
LLMOps / MLOps	MLflow, Prometheus, Grafana, cAdvisor, Node Exporter, ONNX, TensorRT, llama.cpp, NVIDIA Dynamo, vLLM, SGLang, GGUF Quantization (Q4_K_M), Edge Inference Deployment, Langfuse, Opik
Backend	FastAPI, Flask, Django, Node.js, Express.js, SQLAlchemy, Alembic, Prisma
Frontend	React, React Router DOM, TanStack Router, TanStack Query
Databases	PostgreSQL, MySQL, MongoDB

WORK EXPERIENCE

CloudAngles **Aug 2025 – Present**
Software Engineer – Lead, One08labs *Hyderabad*

- Lead engineering for **One08labs**, CloudAngles' AI voice orchestration platform – serving as the primary technical point of contact (SPOC) for enterprise customers, gathering requirements and scoping solutions, and shipping frequent releases so customers get continuous product improvements.
- Engineered the complete telephony infrastructure for the platform, integrating SIP trunking, call routing, Speech-to-Text (STT), LLM inference, and Text-to-Speech (TTS) into a production-ready pipeline supporting reliable inbound and outbound AI voice calls.
- Achieved end-to-end voice conversation latency of approximately 1.15 seconds on Vapi, combining AssemblyAI for transcription, GPT-4.1 for inference, and ElevenLabs for text-to-speech, significantly improving real-time conversational experience.
- Implemented and benchmarked self-hosted LLM inference across vLLM, SGLang, and TensorRT serving backends, integrating NVIDIA Dynamo to disaggregate prefill and decode computation across separate devices – rather than requiring both phases on the same device – improving inference efficiency and speed.
- Architected scalable infrastructure for **mlangles**, CloudAngles' LLMOps platform – including staging environments, model hub, deployment pipelines, and experiment management supporting multiple internal and customer-facing use cases.
- Developed a fine-tuning framework supporting LoRA, QLoRA, Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), Parameter-Efficient Fine-Tuning (PEFT), and Full Fine-Tuning.
- Built automated MLOps pipelines for dataset management, experiment tracking, model versioning, and deployment workflows.
- Developed model optimization pipelines including quantization and runtime conversion using llama.cpp, ONNX, and TensorRT.
- Converted fine-tuned LLMs to Q4_K_M GGUF format using llama.cpp and enabled on-device inference for edge deployment, shipping the capability as a production feature of the platform.
- Implemented evaluation pipelines for RAG applications including hallucination detection, answer relevance, and context scoring.
- Integrated Prometheus, Grafana, cAdvisor, and Node Exporter to enable production monitoring and system observability.

CloudAngles **Mar 2025 – Jul 2025**

GenAI Developer Intern *Hyderabad*

- Developed a centralized system monitoring dashboard for internal AI products.
- Integrated customized computer vision models with Roboflow for dynamic image classification.
- Built AI agents using LangGraph to automate enterprise workflows.
- Developed reusable UI components for chatbot and AI agent creation platforms.

Prathibha Innovations **Dec 2024 – Jan 2025**

SDE Intern *Hyderabad (Remote)*

- Developed a custom user management system using Express.js.
- Configured Prisma ORM for scalable and maintainable database operations.

PROJECTS

Real-Time Voice Agent Orchestration Layer | *One08labs – WebRTC, LiveKit (SIP, Ingress, Egress, Agents)*

- Built an orchestration layer coordinating a cascaded voice pipeline across STT, LLM, and TTS models, exposed over WebRTC to support real-time, low-latency voice interaction.
- Integrated the orchestration layer with SIP to extend real-time voice interaction to traditional telephony calls, leveraging LiveKit's SIP, Ingress, and Egress components alongside its Agents framework.
- Enabled One08labs to support both WebRTC-based and SIP-based voice sessions through a single orchestration layer, forming the core real-time interaction backbone of the platform.

Multi-Backend LLM Inference Optimization | *mlangles – vLLM, SGLang, TensorRT, NVIDIA Dynamo*

- Implemented and evaluated LLM inference across vLLM, SGLang, and TensorRT serving backends within mlangles to identify the most efficient configuration for low-latency production workloads.
- Integrated NVIDIA Dynamo for inference scheduling and orchestration, improving execution efficiency across the platform's serving layer.
- Findings directly informed the serving-stack architecture adopted for mlangles' production inference infrastructure, including downstream real-time workloads such as voice.

Edge LLM Deployment via GGUF Quantization | *mlangles – llama.cpp, GGUF (Q4_K_M)*

- Designed a quantization pipeline converting fine-tuned LLMs to Q4_K_M GGUF format using llama.cpp for lightweight, on-device inference.
- Shipped the capability as a feature within mlangles, enabling model deployment on edge devices without cloud inference dependency.
- Reduced model footprint to support inference on resource-constrained hardware.

AWARDS & HONORS

Gold Medal, State Level Science Talent Search Examination (SLSTE)